

Sec 3.3 – The Product and Quotient Rules

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

In Chap 3, each section introduces a new rule, or shortcut, to differentiation. So as we learn new rules, we still want to be aware of the rules we've learned previously. And we will be able to calculate many of our derivatives multiple ways.

For example, we can do the following derivative with what we learned in the previous sections.

Find $\frac{dy}{dx}$ when $y = x^5(4x^2 + 3x)$ First, we just have to distribute: $y = 4x^7 + 3x^6$

Then the derivative is simply: $\frac{dy}{dx} = 28x^6 + 18x^5$

We will cover Sec 3.3 over two classes. Today, we focus on the Product Rule. It will give us another way to find the derivative above, as well as other functions we wouldn't be able to find the derivative of with what we know so far.

Theorem

If f and g are differentiable functions, then fg is differentiable and

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

$$\text{Or } \frac{d}{dx}(f(x) \cdot g(x)) = \left(\frac{d}{dx}f(x)\right) \cdot g(x) + f(x) \cdot \left(\frac{d}{dx}g(x)\right)$$

So, let's use the Product Rule to find the derivative of the function above, Find $\frac{dy}{dx}$ when $y = x^5(4x^2 + 3x)$

Solution: Using the product rule,

$$\begin{aligned}\frac{d}{dx}x^5(4x^2 + 3x) &= \left(\frac{d}{dx}x^5\right)(4x^2 + 3x) + (x^5)\left(\frac{d}{dx}(4x^2 + 3x)\right) \\ &= (5x^4)(4x^2 + 3x) + x^5(8x + 3).\end{aligned}$$

If we simplify, we get $20x^6 + 15x^5 + 8x^6 + 3x^5 = 28x^6 + 18x^5$, matching our previous answer.

Let's try a few other examples:

It's OK to stop here, no need to simplify!

$$y = (3x^5 + 3x + 7)(x^{-2} + 6x + 3)$$

Solution: Using the product rule,

$$\begin{aligned}y' &= (3x^5 + 3x + 7)'(x^{-2} + 6x + 3) + (3x^5 + 3x + 7)(x^{-2} + 6x + 3)' \\ &= (15x^4 + 3)(x^{-2} + 6x + 3) + (3x^5 + 3x + 7)(-2x^{-3} + 6).\end{aligned}$$

$$y = 4x^2e^x$$

Solution: Using the product rule,

$$y' = (4x^2)'(e^x) + (4x^2)(e^x)' = 8xe^x + 4x^2e^x.$$

And one more:

$$\frac{d}{dx} x^3 2^x = 3x^2 \cdot 2^x + \ln 2 \cdot x^3 \cdot 2^x$$

Partner up

- ① Suppose $f(3) = 2$ and $f'(3) = -4$ and let $F(x) = x^3 f(x)$. Find $F'(3)$.
(Hint: Write out the product rule for $F'(x)$.)

- ② Find the equation of the tangent line to $f(x) = (x^2 + 2x)(3x^4 - 1)$ at $x = 1$.

- ③ Find $\frac{d}{dx} \left[(\sqrt{x} + 2x) \left(\frac{3}{x^3} - 8x^2 \right) \right]$

1. Suppose $f(3) = 2$ and $f'(3) = -4$ and let $F(x) = x^3 f(x)$. Find $F'(3)$.
(Hint: Write out the product rule for $F'(x)$.)

Solution: Taking the derivative using the product rule,

$$F'(x) = (x^3)(f(x))' + (x^3)'f(x) = x^3 f'(x) + 3x^2 f(x)$$

$$\text{Then } F'(3) = 3^3 f'(3) + 3(3)^2 f(3) = 27(-4) + 27(2) = -54.$$

2. Find the equation of the tangent line to $f(x) = (x^2 + 2x)(3x^4 - 1)$ at $x = 1$.

Solution: The tangent line will have form $y = m(x - a) + b$. The x value is 1, so this is a . The y value is $f(1) = (1 + 2)(3 - 1) = 6$. To find m , we need the derivative to find the slope of the tangent line and we find the derivative using the product rule:

$$\begin{aligned} f'(x) &= (x^2 + 2x)(3x^4 - 1)' + (x^2 + 2x)'(3x^4 - 1) \\ &= (x^2 + 2x)(12x^3) + (2x + 2)(3x^4 - 1). \end{aligned}$$

Then the slope m is the derivative at $x = 1$:

$$f'(1) = (1 + 2)(12) + (2 + 2)(3 - 1) = 44.$$

Therefore the equation of the tangent line is $y = 44(x - 1) + 6$.

3. Find $\frac{d}{dx} (\sqrt{x} + 2x) \left(\frac{3}{x^3} - 8x^2 \right)$

Solution: By the product rule,

$$\begin{aligned} &\frac{d}{dx} \left[(\sqrt{x} + 2x) \left(\frac{3}{x^3} - 8x^2 \right) \right] \\ &= (\sqrt{x} + 2x) \frac{d}{dx} \left(\frac{3}{x^3} - 8x^2 \right) + \frac{d}{dx} (\sqrt{x} + 2x) \left(\frac{3}{x^3} - 8x^2 \right) \\ &= (\sqrt{x} + 2x) \left(\frac{-9}{x^4} - 16x \right) + \left(\frac{1}{2\sqrt{x}} + 2 \right) \left(\frac{3}{x^3} - 8x^2 \right) \end{aligned}$$